

## Database Systems Assessment – 5

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```
SQL> CREATE TABLE EMPLOYEES (  
  2  emp_id NUMBER PRIMARY KEY,  
  3  first_name VARCHAR2(20) NOT NULL,  
  4  last_name VARCHAR2(20) NOT NULL,  
  5  job_id VARCHAR2(10) NOT NULL REFERENCES JOBS(job_id),  
  6  salary NUMBER NOT NULL CHECK (salary > 0),  
  7  commission_pct NUMBER CHECK (commission_pct BETWEEN 0 AND 1),  
  8  hire_date DATE DEFAULT SYSDATE NOT NULL,  
  9  dept_id NUMBER REFERENCES DEPARTMENTS(dept_id),  
 10  manager_id NUMBER REFERENCES EMPLOYEES(emp_id)  
 11 );
```

Table created.

```
SQL> CREATE TABLE DEPARTMENTS (  
  2  dept_id NUMBER,  
  3  dept_name VARCHAR2(30) NOT NULL,  
  4  location VARCHAR2(20) NOT NULL,  
  5  CONSTRAINT dept_pk PRIMARY KEY(dept_id),  
  6  CONSTRAINT dept_name_unique UNIQUE(dept_name)  
  7 );
```

Table created.

```
SQL> CREATE TABLE JOBS (  
  2  job_id VARCHAR2(10),  
  3  job_title VARCHAR2(30) NOT NULL,  
  4  min_salary NUMBER NOT NULL,  
  5  max_salary NUMBER NOT NULL,  
  6  CONSTRAINT jobs_pk PRIMARY KEY(job_id),  
  7  CONSTRAINT jobs_min_check CHECK(min_salary >= 0),  
  8  CONSTRAINT jobs_max_check CHECK(max_salary >= min_salary)  
  9 );
```

Table created.

1)

```
SQL> CREATE VIEW DEPT_TOP_EARNERS AS
 2  SELECT e.emp_id, e.first_name, e.last_name,
 3     e.dept_id, e.salary
 4  FROM EMPLOYEES e
 5  WHERE e.salary = (
 6     SELECT MAX(e2.salary)
 7  FROM EMPLOYEES e2
 8     WHERE e2.dept_id = e.dept_id
 9  );
```

View created.

```
SQL> SELECT * FROM DEPT_TOP_EARNERS;
```

| EMP_ID | FIRST_NAME | LAST_NAME | DEPT_ID | SALARY |
|--------|------------|-----------|---------|--------|
| 5      | Kiran      | Das       | 40      | 90000  |
| 7      | Ananya     | Iyer      | 20      | 55000  |
| 4      | Sneha      | Rao       | 10      | 85000  |
| 8      | Rohan      | Mehta     | 30      | 95000  |

```
SQL> SELECT *
 2  FROM DEPT_TOP_EARNERS
 3  WHERE salary > (SELECT AVG(salary) FROM EMPLOYEES);
```

| EMP_ID | FIRST_NAME | LAST_NAME | DEPT_ID | SALARY |
|--------|------------|-----------|---------|--------|
| 5      | Kiran      | Das       | 40      | 90000  |
| 4      | Sneha      | Rao       | 10      | 85000  |
| 8      | Rohan      | Mehta     | 30      | 95000  |

2)

```
SQL> CREATE VIEW DEPT_SALARY_GAP AS
 2 SELECT dept_id,
 3 MAX(salary) - MIN(salary) AS salary_gap
 4 FROM EMPLOYEES
 5 WHERE dept_id IS NOT NULL
 6 GROUP BY dept_id;
```

View created.

```
SQL> SELECT * FROM DEPT_SALARY_GAP;
```

| DEPT_ID | SALARY_GAP |
|---------|------------|
| 10      | 60000      |
| 40      | 0          |
| 20      | 10000      |
| 30      | 20000      |

```
SQL> SELECT *
 2 FROM DEPT_SALARY_GAP
 3 WHERE salary_gap > (
 4 SELECT AVG(salary_gap)
 5 FROM DEPT_SALARY_GAP
 6 );
```

| DEPT_ID | SALARY_GAP |
|---------|------------|
| 10      | 60000      |

3)

```
SQL> CREATE VIEW SALARY_COMPLIANCE AS
 2 SELECT e.emp_id,
 3 e.first_name,
 4 e.last_name,
 5 e.job_id,
 6 e.salary,
 7 j.min_salary,
 8 j.max_salary,
 9 CASE
10 WHEN e.salary BETWEEN j.min_salary AND j.max_salary
11 THEN 'COMPLIANT'
12 ELSE 'VIOLATION'
13 END AS status,
14 CASE
15 WHEN e.salary < j.min_salary THEN j.min_salary - e.salary
16 WHEN e.salary > j.max_salary THEN e.salary - j.max_salary
17 ELSE 0
18 END AS deviation
19 FROM EMPLOYEES e
20 JOIN JOBS j
21 ON e.job_id = j.job_id;
```

View created.

```
SQL> SELECT * FROM SALARY_COMPLIANCE;
```

| EMP_ID     | FIRST_NAME | LAST_NAME | JOB_ID    | SALARY |
|------------|------------|-----------|-----------|--------|
| MIN_SALARY | MAX_SALARY | STATUS    | DEVIATION |        |
| 1          | Arun       | Kumar     | DEV       | 60000  |
| 30000      | 80000      | COMPLIANT | 0         |        |
| 5          | Kiran      | Das       | MGR       | 90000  |
| 50000      | 100000     | COMPLIANT | 0         |        |
| 7          | Ananya     | Iyer      | HR        | 55000  |
| 25000      | 60000      | COMPLIANT | 0         |        |

  

| EMP_ID     | FIRST_NAME | LAST_NAME | JOB_ID    | SALARY |
|------------|------------|-----------|-----------|--------|
| MIN_SALARY | MAX_SALARY | STATUS    | DEVIATION |        |
| 2          | Priya      | Sharma    | DEV       | 75000  |
| 30000      | 80000      | COMPLIANT | 0         |        |
| 3          | Rahul      | Verma     | HR        | 45000  |
| 25000      | 60000      | COMPLIANT | 0         |        |
| 4          | Sneha      | Rao       | DBA       | 85000  |
| 40000      | 90000      | COMPLIANT | 0         |        |

  

| EMP_ID     | FIRST_NAME | LAST_NAME | JOB_ID    | SALARY |
|------------|------------|-----------|-----------|--------|
| MIN_SALARY | MAX_SALARY | STATUS    | DEVIATION |        |
| 6          | Vijay      | Nair      | DEV       | 25000  |
| 30000      | 80000      | VIOLATION | 5000      |        |
| 8          | Rohan      | Mehta     | DBA       | 95000  |
| 40000      | 90000      | VIOLATION | 5000      |        |
| 9          | Meena      | Paul      | DEV       | 50000  |
| 30000      | 80000      | COMPLIANT | 0         |        |

9 rows selected.

```
SQL> SELECT *  
2 FROM SALARY_COMPLIANCE  
3 WHERE status = 'VIOLATION'  
4 ORDER BY deviation DESC;
```

| EMP_ID     | FIRST_NAME | LAST_NAME | JOB_ID    | SALARY |
|------------|------------|-----------|-----------|--------|
| MIN_SALARY | MAX_SALARY | STATUS    | DEVIATION |        |
| 8          | Rohan      | Mehta     | DBA       | 95000  |
| 40000      | 90000      | VIOLATION | 5000      |        |
| 6          | Vijay      | Nair      | DEV       | 25000  |
| 30000      | 80000      | VIOLATION | 5000      |        |

4)

```
SQL> CREATE VIEW LOCATION_SALARY_BILL AS
 2  SELECT d.location,
 3  SUM(e.salary) AS total_salary
 4  FROM EMPLOYEES e
 5  JOIN DEPARTMENTS d
 6  ON e.dept_id = d.dept_id
 7  GROUP BY d.location;
```

View created.

```
SQL> SELECT * FROM LOCATION_SALARY_BILL;
```

| LOCATION  | TOTAL_SALARY |
|-----------|--------------|
| Chennai   | 340000       |
| Mumbai    | 90000        |
| Bangalore | 100000       |

```
SQL> SELECT location,
 2  total_salary,
 3  ROUND(
 4  total_salary * 100 /
 5  (SELECT SUM(total_salary) FROM LOCATION_SALARY_BILL),
 6  2
 7  ) AS percentage
 8  FROM LOCATION_SALARY_BILL
 9  WHERE total_salary = (
10  SELECT MAX(total_salary)
11  FROM LOCATION_SALARY_BILL
12  );
```

| LOCATION | TOTAL_SALARY | PERCENTAGE |
|----------|--------------|------------|
| Chennai  | 340000       | 64.15      |

5)

```
SQL> CREATE VIEW EMP_MANAGER_VIEW AS
 2  SELECT e.emp_id,
 3         e.first_name AS employee_name,
 4         e.salary AS employee_salary,
 5         m.first_name AS manager_name,
 6         m.salary AS manager_salary,
 7         d.dept_name
 8  FROM EMPLOYEES e
 9  JOIN EMPLOYEES m
10  ON e.manager_id = m.emp_id
11  LEFT JOIN DEPARTMENTS d
12  ON e.dept_id = d.dept_id;
```

View created.

```
SQL> SELECT * FROM EMP_MANAGER_VIEW;
```

| EMP_ID | EMPLOYEE_NAME | EMPLOYEE_SALARY | MANAGER_NAME |
|--------|---------------|-----------------|--------------|
| 4      | Sneha         | 85000           | Arun         |
| 60000  | IT            |                 |              |
| 6      | Vijay         | 25000           | Arun         |
| 60000  | IT            |                 |              |
| 3      | Rahul         | 45000           | Ananya       |
| 55000  | HR            |                 |              |

| EMP_ID | EMPLOYEE_NAME | EMPLOYEE_SALARY | MANAGER_NAME |
|--------|---------------|-----------------|--------------|
| 2      | Priya         | 75000           | Arun         |
| 60000  | Finance       |                 |              |
| 8      | Rohan         | 95000           | Priya        |
| 75000  | Finance       |                 |              |
| 9      | Meena         | 50000           | Arun         |
| 60000  |               |                 |              |

6 rows selected.

```
SQL> SELECT *  
  2 FROM EMP_MANAGER_VIEW  
  3 WHERE employee_salary >= manager_salary * 1.20;
```

| EMP_ID | EMPLOYEE_NAME | EMPLOYEE_SALARY | MANAGER_NAME |
|--------|---------------|-----------------|--------------|
| 2      | Priya         | 80000           | Arun         |
| 60000  | Finance       |                 |              |
| 4      | Sneha         | 85000           | Arun         |
| 60000  | IT            |                 |              |